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Underwater image captioning: Challenges, models, and datasets

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ABSTRACT

We delve into the nascent field of underwater image captioning from three perspectives: challenges, models, and datasets. One challenge arises from the disparities between natural images and underwater images, which hinder the use of the former to train models for the latter. Another challenge exists in the limited feature extraction capabilities of current image captioning models, impeding the generation of accurate underwater image captions. The final challenge, albeit not the least significant, revolves around the insufficiency of data available for underwater image captioning. This insufficiency not only complicates the training of models but also poses challenges for evaluating their performance effectively. To address these challenges, we make three novel contributions. First, we employ a physics-based degradation technique to transform natural images into degraded images that closely resemble realistic underwater images. Based on the degraded images, we develop a meta-learning strategy specifically tailored for underwater tasks. Second, we develop an underwater image captioning model based on scene-object feature fusion. It fuses underwater scene features extracted by ResNeXt and object features localized by YOLOv8, yielding comprehensive features for underwater image captioning. Last but not least, we construct an underwater image captioning dataset covering various underwater scenes, with each underwater image annotated with five accurate captions for the purpose of comprehensive training and validation. Experimental results on the new dataset validate the effectiveness of our novel models. The code and datasets are released at https://gitee.com/LHY-CODE/UICM-SOFF.

1. Introduction

Underwater photogrammetry is a significant research area for exploring the underwater world. Notable progress has been made in underwater imaging (Wang et al., 2023, 2024a), underwater 3D reconstruction (Bruno et al., 2011; She et al., 2022), and underwater photogrammetric modeling (Telem and Filin, 2010; Hou et al., 2024a). Among these, underwater imaging related research (Liang et al., 2022; Ren et al., 2024a) has played a crucial role in underwater exploration (e.g., ocean energy exploitation and underwater life monitoring). Improvements in underwater image quality have enhanced our ability to observe the underwater environment. However, undifferentiated processing for the purpose of clear perception has hindered the comprehensive expression of their characteristics. Achieving intelligent perception of underwater images and transitioning from "visual perception" to "semantic understanding" remain a long-term research goal in underwater image photogrammetry (Ren et al., 2024b).

Image captioning technology automatically converts visual content from images into natural and fluent textual information, effectively conveying the semantic information expressed in the image (Vinyals et al., 2015). In the analysis of underwater images, this technology can generate accurate and semantically coherent captions for underwater scenes. This advancement opens the possibility of cognitive perception of the underwater world. Therefore, underwater image captioning technology has demonstrated promising potential in enhancing our understanding of underwater environments.

The remaining sections of this paper are organized as follows: Section 2 reviews the literature on natural image captioning and related datasets. Section 3 discusses three primary challenges in underwater image captioning and outlines our novel contributions. Section 4 presents several novel models, which put together form a new paradigm for underwater image captioning. Section 5 gives a description of the construction and usage of new datasets for this study. Section 6 presents the experiments and analyzes the results. Finally, Section 7 concludes the paper.

2. Related work

In this section, we review the literature on natural image captioning and related datasets.

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2.1. Natural image captioning

Early natural image captioning techniques primarily utilized recurrent neural network (RNN) based encoder–decoder architectures. Vinyals et al. (2015) pioneered the use of convolutional neural networks (CNNs) to extract image features and RNNs to generate captions. Subsequently, attention mechanisms (Xu et al., 2015; Lu et al., 2017; Li et al., 2017) emerged to focus on relevant regions during captioning. Xu et al. (2015) introduced the soft attention mechanism into image captioning models, replacing static global variables with dynamically evolving visual representations and focusing on image regions most relevant to caption generation. Zhu et al. (2018) introduced triple attention, integrating image context into long short-term memory (LSTM) stages. Anderson et al. (2018) combined bottom-up and top-down attention mechanisms to measure focus at the object level. Furthermore, Yao et al. (2018) presented GCN-LSTM, which captures object relationships to improve caption quality. However, RNNs and LSTMs faced criticisms due to their rigidity, limitations in expression, and other complexities. Their recurrent nature often leads to difficulties in memorizing inputs from many steps ago, yielding high-frequency phrase repetitions without regard to visual context.

Inspired by successes in natural language processing (NLP), recent advancements have applied transformer architectures to image captioning. Herdade et al. (2019) introduced object relation transformers tailored for integrating spatial object relationships into caption generation. Cornia et al. (2020) proposed the M2 Transformer, enhancing feature encoding, capturing multi-layered relationships, and decoding diverse features. Further progresses include X-Transformer (Pan et al., 2020), S2 Transformer (Zeng et al., 2022), DeeCap (Fei et al., 2022), DFT (Zhang et al., 2023), and LSTNet (Ma et al., 2023). Furthermore, Zhang et al. (2021) introduced grid augmented and adaptive attention modules to address limitations in visual tasks. These methods typically involve two stages: the first stage uses an offline visual feature extractor to abstract features from images, while the second stage feeds these features into models for generating captions. Hence, these methods are normally referred to as two-stage models. Despite their success, the two-stage models face challenges such as data distribution mismatch and reliance on offline feature extraction, which limit their ability to generalize effectively.

To address these issues, there is growing interest in one-stage models that enable end-to-end caption generation directly from input images. One such model is VC-GPT (Luo et al., 2022), which integrates a CLIP-ViT encoder with a GPT-2 decoder for captioning, eliminating the need for separate object detectors. Another model is PureT (Wang et al., 2022), which employs an end-to-end Transformer model with pre-fusion to enhance cross-modal interaction. Additionally, ViTCAP (Fang et al., 2022) is a detector-free model that utilizes grid representations and CTN within a vision Transformer, offering simplified yet competitive captioning capabilities. The one-stage models reduce intermediate steps, enabling efficient operation. However, they heavily rely on sufficient training data. In this context, the models trained on natural images are challenging to straightforwardly apply to underwater images. Furthermore, most natural image captioning models, no matter one-stage or two-stage, prioritize object features extracted during pre-training but often underestimate the importance of comprehensive scene descriptions, which are crucial for underwater image captioning.

2.2. Related datasets

Another crucial factor for achieving effective image captioning relies on a high-quality image captioning dataset. Such a dataset should include numerous manually annotated diverse captions covering various scenes and objects. This diversity ensures that models can deeply learn and understand the intricate connections between image content and its corresponding captions. Existing image captioning datasets such as Flickr8K (Hodosh et al., 2013), Flickr30K (Plummer et al., 2015), COCO Caption (Chen et al., 2015), Conceptual Captions (Sharma et al., 2018), TextCaps (Sidorov et al., 2020), and the Fashion Captioning Dataset (Yang et al., 2020) primarily focus on daily life natural environments, with few extensions to underwater scenes.

From the research literature reviewed and analyzed above, we have the following observations, which motivate study in this paper. Although effective models have been developed for natural image captioning, they cannot be straightforwardly applied to the underwater image captioning task, which is the primary focus of our study. Significant disparities exist between natural images and underwater images, preventing the settings for natural images from being straightforwardly applied to underwater tasks. Moreover, datasets specifically tailored for underwater image captioning are limited. Datasets such as UIEB (Li et al., 2019), SQUID (Berman et al., 2020), and UID2021 (Hou et al., 2023) have been developed for tasks like underwater image enhancement (Wang et al., 2023; Hou et al., 2024a) and quality assessment (Hou et al., 2024b). For example, Liang et al. (2024) proposed an effective method for underwater image quality improvement through color, detail, and contrast restoration using the UIEB dataset. However, these datasets do not include the necessary annotations for underwater image captioning. Similarly, URPC2021[1] and DUO (Liu et al., 2021) datasets are widely used in underwater object detection (Xu et al., 2021). For example, Shen et al. (2024) developed a multi-information perception attention module that effectively mitigates underwater background interference and improves the detection of underwater objects using the URPC2021 dataset. However, these datasets do not provide scene information necessary for underwater image caption generation. On the other hand, datasets like SUIM (Islam et al., 2020) and UIIS (Lian et al., 2023), which focus on image segmentation, do not provide information suitable for underwater image captioning.

Despite some preliminary studies on underwater image captioning (Tien et al., 2020; Li et al., 2024; Kerai and Khekare, 2024), this research area remains at its early stages, predominantly relying on traditional CNN, R-CNN, and other deep learning architectures, with very limited datasets available. Therefore, developing new underwater image captioning models and constructing baseline benchmark datasets are crucial steps forward.

3. Challenges and our novel contributions

This section provides a summary of the key challenges hindering the progress of the emerging research field of underwater image captioning. Additionally, novel contributions are presented to address these challenges.

3.1. General challenges

As a nascent research topic, underwater image captioning faces various challenges. We outline three key challenges that significantly hinder the progress of research in this field.

Deficiency of Using Natural Images to Train Models for Underwater Images: Existing studies have shown that the performance of image captioning models heavily relies on pre-trained models (Anderson et al., 2018; Jamil et al., 2024). However, most pre-trained models enhance their feature representation capabilities primarily using image data from natural terrestrial environments. In contrast, the underwater environment presents unique characteristics that make underwater images significantly different from those captured in natural environments (Wang et al., 2024c), thereby impeding the accuracy of pre-trained models, which are based on natural images, in recognizing and interpreting underwater images.

[1] https://git.openi.org.cn/OpenOrcinus_orca.

Limited Feature Extraction Capabilities for Underwater Image Captioning: Current image captioning models often prioritize extracting and encoding region-level features to identify crucial information. While these features excel at providing details about objects, they frequently lack the broader context of the scene. In contrast to natural image captioning, which mainly focuses on objects, underwater image captioning requires accurate descriptions of both scenes and objects. The features extracted by current image captioning models may yield incomplete captions for underwater images due to the insufficient consideration of holistic scenes. Therefore, there is a critical need for advanced feature extraction capabilities specifically tailored for underwater image captioning.

Insufficient Data for Underwater Image Captioning: Existing image captioning models heavily rely on sufficient training data primarily consisting of everyday life scenes, which limits their applicability to underwater environments. On the contrary, current underwater image datasets are significantly smaller in scale, posing challenges for effectively training existing models. Moreover, most underwater image datasets prioritize tasks such as image enhancement and segmentation rather than specifically catering to the needs of underwater image captioning. These factors underscore the urgency and challenge of constructing a comprehensive underwater image captioning dataset to train and evaluate underwater image captioning models.

3.2. Our novel contributions

To address the three challenges listed in Section 3.1, we make three novel contributions.

A Novel Meta-learning Strategy for Bridging Natural and Underwater Image Domains: We develop a novel meta-learning strategy to bridge the gap between natural and underwater image domains. Our strategy leverages an advanced physics-based degradation technique to transform natural images into degraded images that closely resemble realistic underwater scenes. A visual feature extractor is pre-trained using these degraded images to learn underwater features that can handle new underwater tasks. The adaptation of the visual feature extractor from natural images to underwater tasks is enabled by such meta-learning strategy.

A Novel Underwater Image Captioning Model for Enhancing Feature Representation: We develop a novel underwater image captioning model based on scene-object feature fusion, aiming to improve the limited feature extraction capability of existing image captioning models. Specifically, we employ a visual feature extractor of ResNeXt (Xie et al., 2017) to extract scene features from underwater images, effectively capturing their extensive global visual information. Additionally, a high-performance YOLOv8 (Solawetz and Francesco, 2023) object detector is incorporated to accurately localize features for key objects, thereby focusing on salient details. Moreover, a scene-object feature fusion-based Transformer encoder is proposed to fuse the scene features and object features, yielding comprehensive features for underwater image captioning.

A New Benchmark Dataset for Underwater Image Captioning: We construct an underwater image captioning dataset to address the insufficiency of data in the underwater image captioning task. This dataset encompasses various underwater scenes and objects, such as coral reef, shipwrecks, marine animal, underwater vegetation, diving equipment, and artificial underwater structures. It serves as a robust foundation for scene analysis in underwater environments. Each image in the dataset is paired with five captions, ensuring accuracy in depicting underwater scenes. This dataset establishes a new benchmark for both training and assessing underwater image captioning models.

4. Models

This section introduces a meta-learning strategy based on degraded images, a novel underwater image captioning model, and the associated training strategies for these models.

4.1. A meta-learning strategy based on degraded images

We propose a meta-learning strategy that leverages degraded images to achieve outstanding performance in feature extraction across various tasks. One goal of meta-learning is to enable models to learn generalized feature representations that can efficiently adapt to and enhance performance on new tasks. In this context, we describe how we follow the meta-learning principle to pre-train a model for extracting underwater feature representations from virtual underwater images, which are degraded versions of natural images. This strategy enhances the model's ability to handle underwater tasks.

Specifically, we adopt an advanced physics-based degradation technique of submergence generative adversarial network (submergence GAN) (Wang et al., 2024b) to transform natural images into degraded images that closely resemble realistic underwater scenes. Subsequently, these degraded images are used to pre-train a ResNeXt model, which is initialized on the ImageNet dataset (Deng et al., 2009). Firstly, we employ the AdaBins monocular depth estimation network (Bhat et al., 2021) to convert a natural image into a depth map. Next, both the natural image and its corresponding depth map are input into the submergence GAN, which transforms the natural image into a degraded image that closely resembles a realistic underwater image. Subsequently, the degraded image $\mathcal{I}_{\mathrm{d}} \in \mathbb{R}^{C\times W\times H}$ is input into the visual feature extractor of the ResNeXt to obtain scene features $\mathbf{F}_{\mathrm{s}} \in \mathbb{R}^{C'\times W'\times H'}$ that capture global information of the image. Here, C, W, and H represent the number of channels, width, and height of the input degraded image, and C', W', and H' denote the number of channels, width, and height of the output scene features, respectively. Finally, the scene features $\mathbf{F}_{\mathrm{s}}$ are input into the classifier of the ResNeXt to obtain the predicted class probabilities P.

The pre-training of the ResNeXt on degraded images involves comparing the predicted class probabilities P with the true one-hot class labels Y, as detailed in Section 4.3. We refer to the visual feature extractor of ResNeXt, pre-trained on the degraded images, as Meta-ResNeXtor. The parameter values of the Meta-ResNeXtor obtained by the above training process are referred to as meta-weights.

In summary, the Meta-ResNeXtor pre-trained according to the meta-learning strategy, as illustrated in Fig. 1, effectively overcomes the limitations of applying models trained on natural images to underwater tasks. It significantly bridges the gap between natural and underwater image domains, allowing it to efficiently adapt to new underwater tasks. This meta-learning strategy particularly enhances the performance of the subsequent task of underwater image captioning, which is the major focus of this study.

4.2. Underwater image captioning based on scene-object feature fusion

We develop an underwater image captioning model based on scene-object feature fusion (UICM-SOFF) to improve the limited feature extraction capability of existing image captioning models, as illustrated in Fig. 2. Initially, an underwater image is processed using Meta-ResNeXtor, which is obtained following the meta-learning strategy presented in Section 4.1, and YOLOv8, producing scene features and object detection results, respectively. Subsequently, object features are localized in the scene features subject to the object detection results. Next, we develop a scene-object feature fusion-based Transformer encoder to fuse the scene and the object features. Finally, the scene-object fusion features generated by the encoder are fed into a Transformer decoder, which generates a caption for the underwater image. Detailed descriptions of each module are provided in the subsequent paragraphs of this subsection.

Object Feature Localization: We leverage the object detection results from YOLOv8, a state-of-the-art model renowned for its superior performance in detecting objects of various sizes, to accurately localize features for key objects and emphasize salient details, as illustrated in Fig. 3.

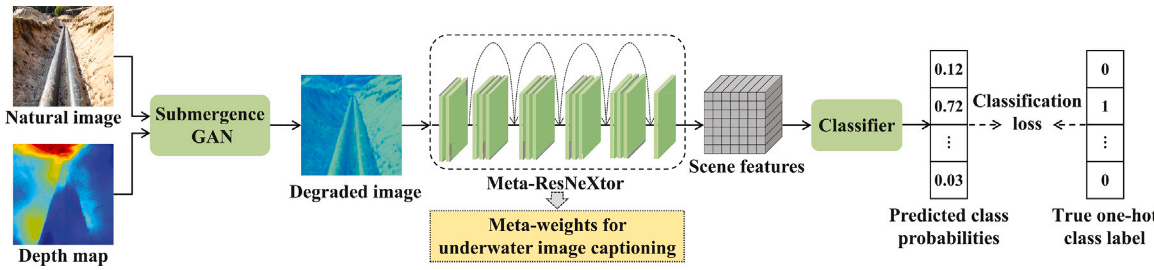


Fig. 1. A meta-learning strategy for enhancing Meta-ResNeXtor for underwater tasks. Solid arrows indicate the implementation process, dashed arrows indicate the training process, and the downwards double arrow presents the parameter values (i.e., meta-weights) involved in meta-learning.

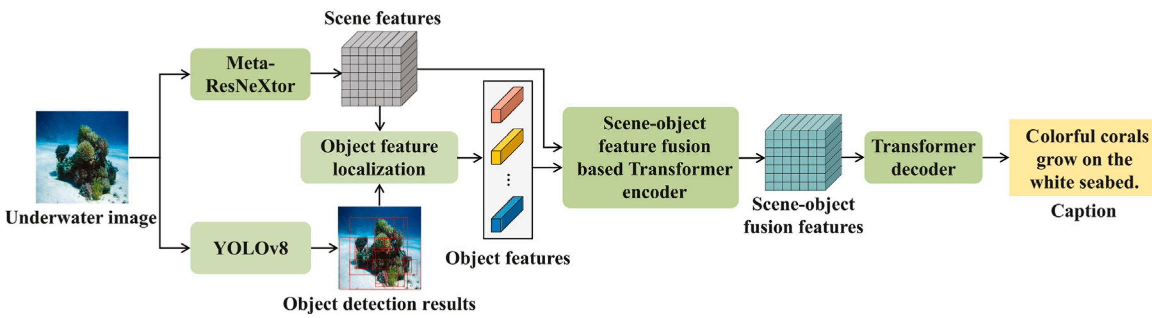


Fig. 2. Underwater image captioning based on scene-object feature fusion.

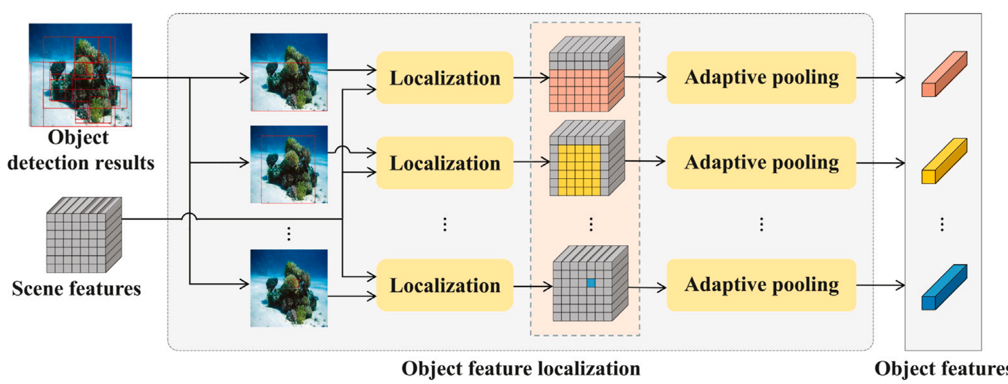


Fig. 3. Object feature localization.

Firstly, an underwater image $\mathcal{I}$ is processed by Meta-ResNeXtor and YOLOv8, yielding scene features $\mathbf{F}_s$ and object detection results $\mathcal{R}_d$, respectively. The object detection results $\mathcal{R}_d$ include essential attributes such as central coordinates, width, and height for each detected object. Subsequently, the object detection results $\mathcal{R}_d$ are sorted in descending order based on the area of the detected objects (i.e., the product of width and height), producing prioritized object detection results $\mathcal{R}$. This prioritization emphasizes objects with larger areas, thereby enhancing focus on salient objects in underwater image captioning tasks.

Then, by utilizing the scene features $\mathbf{F}_s$ and prioritized object detection results $\mathcal{R}$, each detected object is localized to extract its coordinate information within the scene features. Specifically, stride sizes in the horizontal and vertical directions are calculated as $\text{stride}_w = \frac{W}{W'}$ and $\text{stride}_h = \frac{H}{H'}$, respectively. Based on the stride parameters stride_w and stride_h, we employ a coordinate-based localization approach to position the prioritized object detection results $\mathcal{R}$ within the scene features $\mathbf{F}_s$, thereby computing the top-left and bottom-right coordinates for each detected object within the scene features. These position coordinates are then added to the set $\mathcal{A}$. To ensure that no duplicate positions exist and to prevent redundant information from affecting the underwater image captioning model, a deduplication filtering step is applied. This step removes any repeated position coordinates, yielding a set $\mathcal{A}$ that contains only unique entries.

Finally, based on the positional coordinates within the scene features $\mathbf{F}_s$ where each object in set $\mathcal{A}$ is located, adaptive pooling is performed to extract object features $\mathbf{F}_o \in \mathbb{R}^{T\times C}$, where C is the number of feature channels, and T is a threshold introduced to limit the maximum number of detected objects participating in subsequent processing. This measure helps control computational complexity by focusing on the most key objects.

The crux of this methodology hinges on the astute amalgamation of scene features and key object information. This integration guarantees the retention of important underwater image details and simplifies computational complexity, thereby enhancing processing efficiency. This method achieves accurate and efficient extraction of key object features within underwater images.

Scene-Object Feature Fusion Based Transformer Encoder: Unlike natural image captioning, which primarily emphasizes objects, underwater image captioning requires precise descriptions of both overall scenes and key objects. Existing image captioning models often generate incomplete captions for underwater images due to inadequate consideration of holistic scenes. To address these problems, we propose a scene-object feature fusion-based transformer encoder to produce

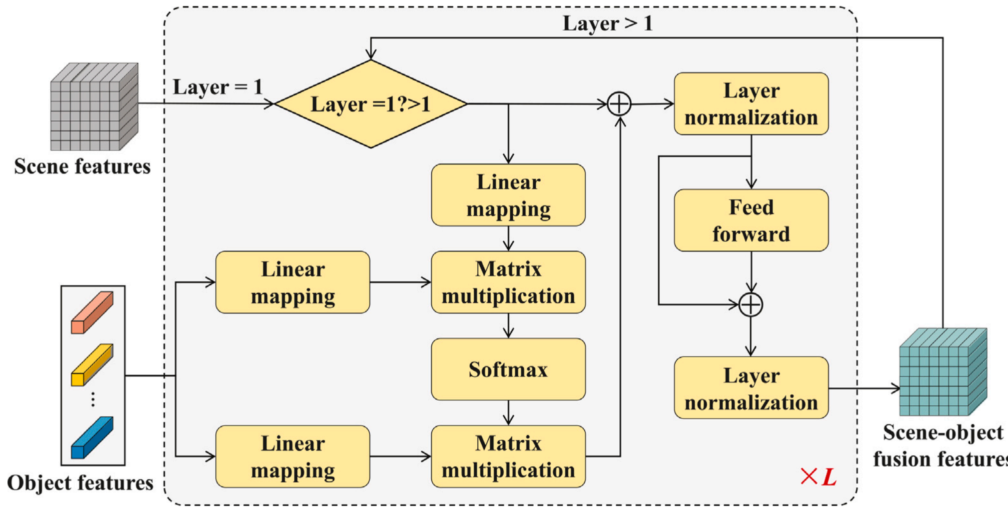


Fig. 4. Scene-object feature fusion based transformer encoder.

comprehensive features for underwater image captioning by leveraging the complementary strengths of scene features $\mathbf{F}_{\mathrm{s}}$ and object features $\mathbf{F}_{\mathrm{o}}$, as illustrated in Fig. 4.

Firstly, the scene features $\mathbf{F}_{\mathrm{s}}$ are mapped into queries $\mathbf{Q}_{\mathrm{s}}$ using linear layers, and the object features $\mathbf{F}_{\mathrm{o}}$ are mapped into keys $\mathbf{K}_{\mathrm{o}}$ and values $\mathbf{V}_{\mathrm{o}}$ as follows:

$$\begin{aligned} \mathbf{Q}_{\mathrm{s}} &= \mathbf{F}_{\mathrm{s}}\mathbf{W}_{\mathrm{q}} + \mathbf{b}_{\mathrm{q}} \\ \mathbf{K}_{\mathrm{o}} &= \mathbf{F}_{\mathrm{o}}\mathbf{W}_{\mathrm{k}} + \mathbf{b}_{\mathrm{k}} \\ \mathbf{V}_{\mathrm{o}} &= \mathbf{F}_{\mathrm{o}}\mathbf{W}_{\mathrm{v}} + \mathbf{b}_{\mathrm{v}}. \end{aligned} \tag{1}$$

where $\mathbf{W}_{\mathrm{q}}$, $\mathbf{W}_{\mathrm{k}}$, and $\mathbf{W}_{\mathrm{v}}$ are learning parameters, $\mathbf{b}_{\mathrm{q}}$, $\mathbf{b}_{\mathrm{k}}$, and $\mathbf{b}_{\mathrm{v}}$ are bias.

Then, the query $\mathbf{Q}_{\mathrm{s}}$ and the key $\mathbf{K}_{\mathrm{o}}$ are utilized to compute attention weights, which enhance the importance of each object relative to each scene feature. These attention weights are applied through matrix multiplication to the value $\mathbf{V}_{\mathrm{o}}$ to obtain the final weighted output $\mathbf{V}_{\mathrm{w}}$:

$$\mathbf{V}_{\mathrm{w}} = \mathrm{softmax}\left(\frac{\mathbf{Q}_{\mathrm{s}}\mathbf{K}_{\mathrm{o}}^{\top}}{\sqrt{d_k}}\right)\mathbf{V}_{\mathrm{o}}, \tag{2}$$

where d_k is the dimension of the key $\mathbf{K}_{\mathrm{o}}$ and $\mathrm{softmax}(\cdot)$ is used to normalize the attention weights, ensuring they sum to 1.

Next, the weighted output $\mathbf{V}_{\mathrm{w}}$, after undergoing residual connections and layer normalization, is input into a feed-forward neural network. This network consists of two fully connected layers and employs a ReLU activation function along with residual connections to facilitate nonlinear transformation. The resulting output is then added back to the original weighted output $\mathbf{V}_{\mathrm{w}}$. The sum is then passed through layer normalization to generate scene-object fusion features $\mathbf{F}_{\mathrm{so}}$. This process is represented as:

$$\mathbf{F}_{\mathrm{so}} = \mathrm{LayerNorm}\left(\mathbf{V}_{\mathrm{w}} + \mathrm{FF}\left(\mathbf{V}_{\mathrm{w}}\right)\right), \tag{3}$$

where $\mathrm{LayerNorm}(\cdot)$ is the layer normalization operation, and $\mathrm{FF}(\cdot)$ denotes the feed-forward neural network.

Finally, the scene-object feature fusion encoder effectively fuses scene features $\mathbf{F}_{\mathrm{s}}$ and object features $\mathbf{F}_{\mathrm{o}}$, thereby enhancing deep-level visual representations through progressively stacked encoder layers, as illustrated in Fig. 4. In the initial encoder layer (i.e., layer = 1), the original scene features $\mathbf{F}_{\mathrm{s}}$ and object features $\mathbf{F}_{\mathrm{o}}$ are directly utilized as inputs. In subsequent encoder layers (i.e., layer > 1), the scene features $\mathbf{F}_{\mathrm{s}}$ are replaced by scene-object fusion features $\mathbf{F}_{\mathrm{so}}$ from the previous encoder layer, preserving the same dimensionality. The encoder progressively refines the feature representations, facilitating the effective capture of complex interdependencies between the underwater scene and objects, thereby improving the performance of the underwater image captioning model.

Transformer Decoder: The highly optimized scene-object fusion features $\mathbf{F}_{\mathrm{so}}$ are input into a standard Transformer decoder (Vaswani, 2017) to generate the predicted underwater image caption $\hat{S}$ as follows:

$$\hat{S} = \mathrm{Transformer}_{\mathrm{decoder}}(\mathbf{F}_{\mathrm{so}}). \tag{4}$$

In summary, this subsection proposes an underwater image captioning model based on scene-object feature fusion. We employ a Meta-ResNeXtor to extract scene features from underwater images, effectively capturing their extensive global visual information. Additionally, we utilize the YOLOv8 object detector to precisely localize key objects, emphasizing crucial details. Furthermore, a scene-object feature fusion encoder is proposed to fuse scene and object features, enhancing the model's capability to represent these features and enabling accurate caption generation for underwater images.

4.3. Training strategies for underwater image captioning models

This subsection introduces the training strategies specifically employed for underwater image captioning models as follows:

Pre-training the ResNeXt model: We utilize the degraded image dataset $\mathcal{D}_{\mathrm{d}} = \{(\mathcal{I}_{\mathrm{d}}^{(n)}, \mathcal{Y}^{(n)})\}_{n=1}^{N}$ to pre-train the ResNeXt model initialized based on the ImageNet dataset, where N represents the total number of samples in the degraded image dataset. The details of the degraded image dataset used in our study will be presented in Section 5.1. Each training data sample consists of a degraded image $\mathcal{I}_{\mathrm{d}}^{(n)}$ and its corresponding true one-hot class label $\mathcal{Y}^{(n)}$. During training, our focus is specifically on optimizing the convolutional layers of the model. Our model enables the model to relearn and adapt to the unique color characteristics and low-level visual information inherent to underwater images (i.e., degraded images), while preserving the excellent performance of its original model. The cross-entropy loss $\mathcal{L}_{\mathrm{CE}}$ for pre-training ResNeXt based on degraded images is given as follows:

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{B}\sum_{b=1}^{B}\sum_{m=1}^{M} Y_m^{(b)} \ln P_m^{(b)}, \tag{5}$$

where B is the batch size, M is the number of classes in the degraded image dataset, $Y_m^{(b)}$ is the mth element of $\mathcal{Y}^{(b)}$ representing the true label for the bth degraded image for the mth class (which is 1 if the degraded image belongs to class m and 0 otherwise), and $P_m^{(b)}$ is the probability of the bth degraded image predicted to belong to class m.

After training, we obtain the visual feature extractor, i.e., Meta-ResNeXtor, along with its corresponding parameter values, i.e., the meta-weights, for the subsequent underwater image captioning task.

Training the UICM-SOFF: We employ the Meta-ResNeXtor configured with meta-weights as the initial visual feature extractor for training the UICM-SOFF. We construct an underwater image captioning dataset $\mathcal{D} = \{(\mathcal{I}^{(j)}, \mathcal{S}^{(j)})\}_{j=1}^{J}$ to train the UICM-SOFF, where J represents the total number of samples in the dataset. The details of the underwater image captioning dataset in our study will be presented in Section 5.2. Each training data sample consists of an underwater image $\mathcal{I}^{(j)}$ and its corresponding ground-truth reference captions $\mathcal{S}^{(j)}$. The training typically begins with supervised learning, followed by reinforcement learning.

In the supervised learning phase, the training loss $\mathcal{L}_{\mathrm{SL}}$ for UICM-SOFF on the underwater image captioning dataset is given by:

$$\mathcal{L}_{\mathrm{SL}} = -\frac{1}{B}\sum_{b=1}^{B}\sum_{l=1}^{L}\sum_{v=1}^{V} s_{l,v}^{(b)} \ln \hat{s}_{l,v}^{(b)}, \tag{6}$$

where B denotes the batch size, L represents the length of the underwater image caption, and V is the vocabulary size. At step l, $s_{l,v}^{(b)}$ is an indicator for the vth word in the vocabulary appearing in the ground-truth reference caption $\mathcal{S}^{(b)}$, and $\hat{s}_{l,v}^{(b)}$ is the predicted probability of the vth word in the vocabulary appearing in the generated caption $\hat{\mathcal{S}}^{(b)}$.

In the reinforcement learning phase, it is a common practice to use the CIDEr score (Vedantam et al., 2015) as the reward. The goal of training is to minimize the negative reward based on self-critical sequence training (SCST) (Rennie et al., 2017). The training loss $\mathcal{L}_{\mathrm{RL}}$ for UICM-SOFF on the underwater image captioning dataset is given by:

$$\mathcal{L}_{\mathrm{RL}} = -\frac{1}{BK}\sum_{b=1}^{B}\sum_{k=1}^{K}\left[r(\hat{\mathcal{S}}_{k}^{(b)}, \mathcal{S}^{(b)}) - \bar{r}^{(b)}\right] \ln p(\hat{\mathcal{S}}_{k}^{(b)}), \tag{7}$$

where $\hat{\mathcal{S}}_{k}^{(b)}$ represents the generated caption for the kth beam search in the bth batch for the underwater image, $r(\hat{\mathcal{S}}_{k}^{(b)}, \mathcal{S}^{(b)})$ is the reward based on the comparison between the generated caption $\hat{\mathcal{S}}_{k}^{(b)}$ and the ground-truth reference caption $\mathcal{S}^{(b)}$, $\bar{r}^{(b)}$ is the average reward across all beams in the bth batch, which is used to normalize the reward scores, and $p(\hat{\mathcal{S}}_{k}^{(b)})$ refers to the probability of the kth beam's generated caption.

In summary, we first pre-train ResNeXt using the degraded image dataset and utilize the trained Meta-ResNeXtor as the visual feature extractor for UICM-SOFF. Then, we adopt a phased training method to train UICM-SOFF using the underwater image captioning dataset to generate accurate captions for underwater images.

5. Datasets

This section details the construction of the degraded image dataset, the underwater image captioning dataset, and their usage.

5.1. Degraded image dataset

We employ a submergence GAN that takes natural images and their corresponding depth maps as inputs, transforming the natural images into degraded images that closely resemble realistic underwater images. The submergence GAN comprises three independent generators, each built upon a ResNet18. These generators are referred to as red, green and blue generators and are specifically used to process the red, green and blue channels of an image to synthesize a degraded image. This setting ensures that the generated images undergo changes only in terms of color and intensity, aligning them with the characteristics of the underwater environment while preserving the original scene structure.

Specifically, we construct a degraded image dataset using ImageNet-100.[2] We select 135 thousand natural images from 100 classes in ImageNet-100. These images along with their corresponding depth images are processed by the submergence GANs in six different color domains (i.e., green, yellow–green, blue–green, blue, dark blue, and purple), which are pre-set in the submergence GANs model (Wang et al., 2024b). This process results in 810,000 degraded images, some examples of which are illustrated in Fig. 5. For each natural image, we randomly select one of the six different color-degraded images generated, and together with the original natural image, these form our degraded image dataset. The training set comprises 260 thousand images (2600 images per class), and the validation set includes 10 thousand images (100 images per class).

[2] https://aistudio.baidu.com/datasetdetail/208729.

Table 1
Main categories of underwater images and their corresponding quantities in the underwater image captioning dataset.

Category	Quantity	Category	Quantity
Fish	1861	Coral reefs	1587
Divers	850	Aquatic plants	253
Sea turtles	228	Artificial objects	366
Cetaceans	117	Other creatures	112
Submersibles	95	Natural environments	896

5.2. Underwater image captioning dataset

We systematically collect 3176 high-quality underwater images from the internet. The underwater images exhibit a broad range of resolutions, with the maximum resolution being 8432 × 4743 pixels and the minimum resolution being 199 × 300 pixels. For consistency and ease of use, all images have been resized to a resolution of 224 × 224 pixels. Both the original images and the resized ones are provided in the code repository, which has been presented in the abstract, to facilitate various tasks. To ensure dataset representativeness, we incorporate diverse underwater images, including distant small targets, dense targets, visual blur, and low lighting, to comprehensively represent real-world underwater conditions. Table 1 provides a detailed summary of the ten distinct categories included in the collected underwater images and their respective quantities. Fish, coral reefs, and divers are among the top three categories represented in the dataset.

The underwater image captions in the dataset are annotated by five volunteers with domain expertise and experience. Each underwater image is annotated with five distinct captions to ensure vocabulary diversity. As a result, the dataset includes 15,880 underwater image captions. We follow specific guidelines during caption construction to ensure fluency and academic quality:

(a) Each underwater image caption should contain at least five words.

(b) Morphological features, color attributes, quantity information, and motion state of all prominent objects in the underwater images are accurately described.

(c) Additional information, such as the behavior of the target objects, background details, and interactions with other organisms, is encouraged to be provided.

(d) Ambiguous words and syntax should not be used.

(e) Fresh vocabulary and flexible syntax are encouraged.

Table 2 shows six exemplary underwater images along with their corresponding detailed ground-truth reference captions. The captions provided by professionals are accurate, vivid, rich in vocabulary, and diverse in syntactic structure, effectively conveying the characteristics and scene information depicted in the underwater images.

We use the natural language processing tool spaCy[3] to analyze the constructed underwater image caption dataset, and the results show that the dataset contains 1469 unique words, with a total vocabulary count of 206,534. Table 3 shows counts and percentages of words belonging to different parts of speech in this dataset, including nouns

[3] https://spacy.io.

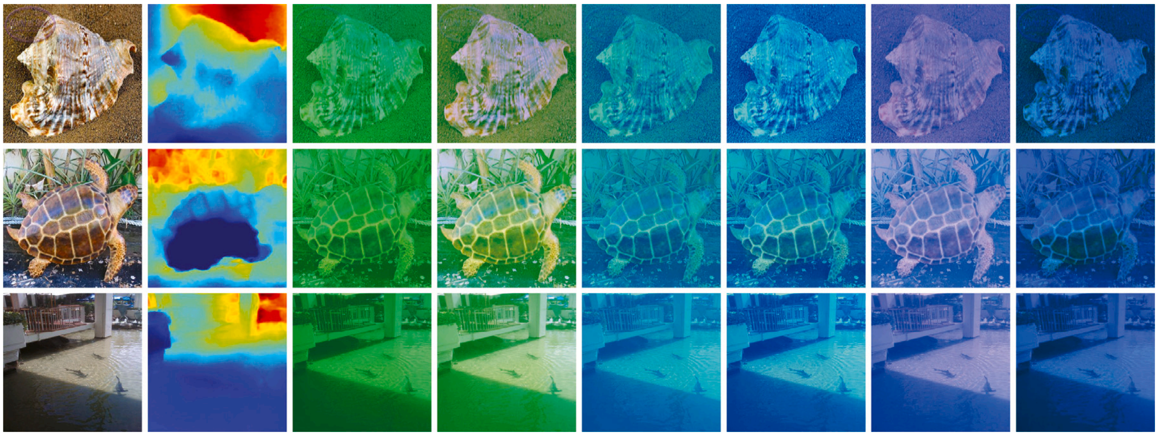
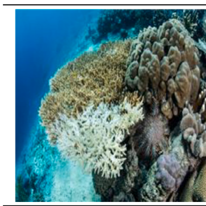
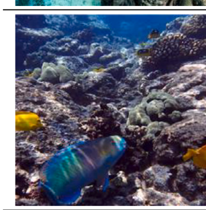
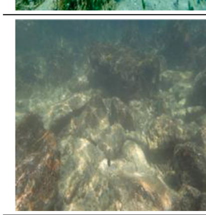
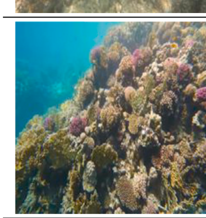
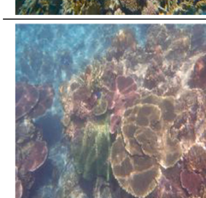
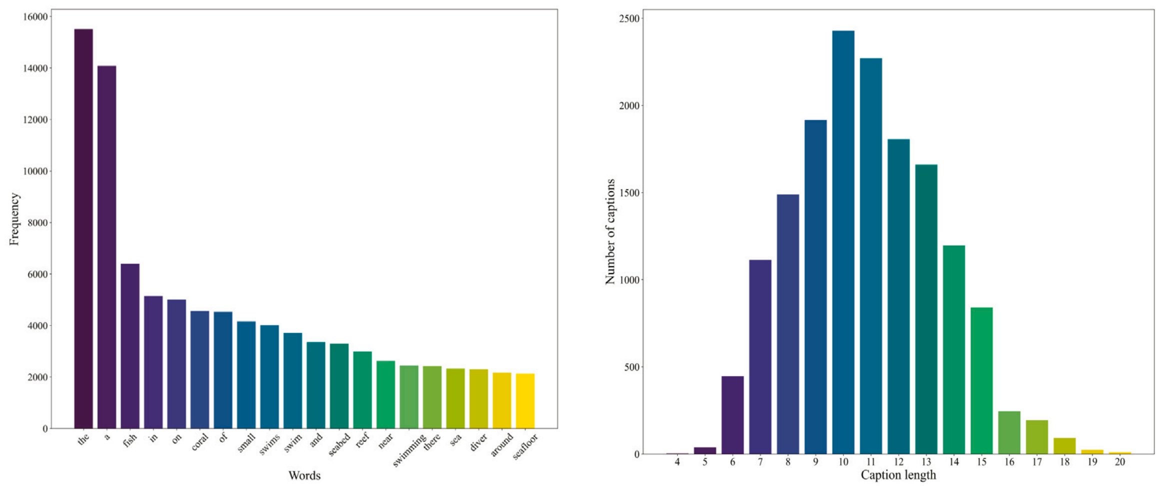


Fig. 5. Examples of degraded images, from left to right: the original image, its depth map, and the corresponding degraded images in green, yellow–green, blue–green, blue, dark blue, and purple. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
Examples from the underwater image captioning dataset.

Underwater image	Ground truth reference captions for underwater image
	GT-1: Coral reefs in seawater come in a variety of colors, such as dark-brown, light-brown, and white. GT-2: An underwater coral reef full of life, with all kinds of corals growing freely. GT-3: Coral reefs in the water grow corals of various shapes and colors. GT-4: An undersea coral reef with corals of various colors and shapes. GT-5: There are many corals of various shapes and colors growing on the underwater reefs.
	GT-1: Several small fish swim and play around the coral reef on the seafloor. GT-2: There are some fish on the seabed, among which a blue and green fish is particularly eye-catching. GT-3: A colorful small fish swims towards the reef at the seafloor. GT-4: There are some corals growing on the seabed, and some small fish swim near them. GT-5: Several small fish swim near the rocks on the seafloor.
	GT-1: A pipe is laid on the seabed and covered with fine sand. GT-2: There are many seagrasses growing near the pipelines laid on the seabed. GT-3: A long pipe is laid on the seabed. GT-4: A pipe is laid on the seabed and covered with fine sand. GT-5: There are many green seagrasses growing on the seabed, and a pipeline is also laid.
	GT-1: The clear water has many rocks of different shapes. GT-2: There are rocks of different shapes in the seawater. GT-3: There are many rocks of different sizes on the seabed. GT-4: There are many rocks of different sizes and shapes on the seabed. GT-5: An underwater scene with rocks of various sizes and shapes.
	GT-1: Corals with different shapes and colors grow on the reef under the sea. GT-2: Colorful corals grow on the reef at the seafloor. GT-3: Many corals grow on the reef at the seafloor. GT-4: Corals with different shapes and colors grow on the seabed. GT-5: Colorful corals grow on the seafloor.
	GT-1: Several small fish swim near the coral. GT-2: Corals of various colors grow on the reefs at the seafloor. GT-3: A clear underwater landscape with a variety of brightly colored corals. GT-4: Corals on the seabed come in various shapes and sizes, some fan-shaped and some disc-shape. GT-5: There are various colorful corals on the seabed, including brown–red, light brown, and green.



(a) Frequency distribution of the top 20 words. (b) Distribution of caption lengths.

Fig. 6. Distribution of the top 20 words and caption lengths in the underwater image captioning dataset.

Table 3
Counts and percentages of words belonging to different parts of speech in the underwater image caption dataset.

	NOUN	ADJ	VERB	ADP	DET	ADV	NUM
Count	60,775	30,671	21,296	28,152	31,908	5651	2238
Percentage	29.43%	14.85%	10.31%	13.63%	15.45%	2.49%	1.08%

(NOUN), adjectives (ADJ), verbs (VERB), adpositions (ADP), determiners (DET), adverbs (ADV), and numerals (NUM). The analysis results indicate that verbs and adjectives make up a significant proportion of captions in the dataset, primarily because the captions focus on depicting various underwater objects, scenes, and their dynamic states.

Fig. 6 presents some basic information about the annotated captions. Fig. 6(a) illustrates the distribution characteristics of the top 20 high-frequency words in the underwater image captioning dataset, while Fig. 6(b) provides a detailed display of the distribution of caption lengths. The distribution exhibits typical characteristics of a normal distribution, with captions containing 6 to 15 words being the most common.

In summary, we present a new benchmark dataset for underwater image captioning encompassing various underwater scenes and objects. Each underwater image in the dataset is accompanied by five accurately annotated captions, addressing the data insufficiency in the underwater image captioning task.

5.3. Usage of the two datasets

The datasets presented in Sections 5.1 and 5.2 are released at the link given in the abstract for public use. In our work, as discussed in Section 4.3, we utilize both the degraded image dataset and the underwater image captioning dataset to train our proposed models for underwater image captioning.

Initially, we utilize the degraded image dataset to pre-train ResNeXt, ensuring that the resulting Meta-ResNeXtor retains its feature extraction capability across various scenes and effectively adapts to the distinctive conditions of underwater environments.

Subsequently, we utilize the underwater image captioning dataset to train UICM-SOFF and conduct two-phase training. In the supervised training phase, the model adjusts parameters by comparing predicted captions with ground-truth reference captions to minimize the loss function. In the reinforcement learning phase, the model is incentivized to generate higher-quality captions through a reward mechanism, encouraging continuous optimization of its generation strategy to improve caption accuracy and fluency.

6. Experiments

In this section, extensive experiments are conducted on the proposed underwater image captioning dataset to demonstrate the effectiveness of UICM-SOFF. First, evaluation metrics and implementation details are introduced. Then, comparison and ablation experiments are conducted and analyzed. Finally, visualization results from our experiments are also provided.

6.1. Experimental settings

Evaluation Metrics: This paper employs six widely used standard metrics to evaluate underwater image captioning performance: BLEU (Papineni et al., 2002), METEOR (Banerjee and Lavie, 2005), ROUGE-L (Lin, 2004), CIDEr (Vedantam et al., 2015), SPICE (Anderson et al., 2016), and S_m (Yang et al., 2024). These metrics collectively gauge the quality of automatically generated underwater image captions. For any given underwater image, a ground-truth reference caption is utilized to compare against the model-generated candidate caption. The definitions of these six metrics are as follows:

(a) BLEU assesses the similarity between the candidate caption and the reference caption by comparing the co-occurrence of n-grams, aiming to minimize redundancy.

(b) METEOR measures the harmonic mean of precision and recall between the candidate caption and the reference caption.

(c) ROUGE-L calculates the similarity between the candidate caption and the reference caption by computing the F-measure based on the longest common subsequence, aiming to effectively capture overlapping information.

(d) CIDEr treats each caption as a document and computes the cosine similarity of TF–IDF vectors to assess the similarity between the candidate caption and the reference caption. It then aggregates the similarity of n-grams of varying lengths to derive the final score.

(e) SPICE emphasizes semantic content in human evaluation by using graph-based semantic representations to encode objects, attributes, and relationships in captions. It calculates F-scores based on these semantic aspects.

(f) S_m represents the average score of four evaluation metrics: BLEU-4, METEOR, ROUGE-L, and CIDEr. It can comprehensively evaluate the quality of generated underwater image captions.

The scores of BLEU, METEOR, ROUGE-L, and SPICE range from 0 to 1, while the CIDEr score ranges from 0 to 5, and S_m ranges from 0 to 2. Higher scores on these metrics indicate a strong resemblance between predicted and human-annotated reference captions.

Implementation Details: In this study, we randomly divide the underwater image captioning dataset into training, validation, and testing sets, with proportions of 70%, 15%, and 15%, respectively. Experiments are conducted on a workstation featuring an Intel(R) Xeon(R) Silver 4214R CPU @ 2.40 GHz processor, 128 GB of RAM, and an NVIDIA GeForce RTX 2080 Ti GPU. The operating system used is Ubuntu 20.04, with PyTorch 1.7.1 employed as the deep learning framework and CUDA 10.1 utilized for GPU acceleration.

We utilize Meta-ResNeXt pre-trained on a degraded image dataset as the visual feature extractor for UICM-SOFF, with the input image size set to 224 × 224 pixels. We train the Meta-ResNeXt model using a stochastic gradient descent optimizer with an initial learning rate of 0.1, momentum of 0.9, weight decay set to 1×10^{-4}, and a training batch size of 20. Both our scene-object feature fusion encoder and standard Transformer-based decoder have three layers, with eight self-attention heads per layer, and an inner dimension of the feed-forward neural network set to 2048. The YOLOv8 model uses pre-trained weights obtained from the COCO dataset. Additionally, within UICM-SOFF, the threshold T for the number of objects is set to 30. We employ the Adam optimizer for parameter adjustment and set the batch size to 20. In the supervised learning phase, we set the minimum epoch to 15 and employ an epoch decay schedule to adjust the learning rate, following RSTNet (Zhang et al., 2021). In the reinforcement learning phase, the learning rate is fixed at 5×10^{-6}. During validation and testing, the beam size of the beam search is set to 5.

6.2. Comparison with existing image captioning models

To illustrate the effectiveness of UICM-SOFF, several existing image captioning models are compared with UICM-SOFF on the underwater image captioning dataset, including state–of–the–art attention-based and Transformer-based models.

(a) NIC (Vinyals et al., 2015) utilizes CNNs to extract image features and combines RNNs to generate captions.

(b) Soft-Att (Xu et al., 2015) combines CNNs and LSTM, enhancing model performance by introducing a soft attention mechanism.

(c) Bi-LSTM (Wang et al., 2016) uses deep bidirectional LSTMs to generate image captions by integrating both visual features and context information effectively.

(d) Spatial-Att (Lu et al., 2017) allocates spatial weights to focus on local regions crucial for generating captions, enhancing the model's ability to comprehend image details.

(e) Adaptive-Att (Lu et al., 2017) introduces adaptive adjustment features, enabling the model to intelligently select fixation points based on the current caption generation state, facilitating dynamic capture and utilization of image information.

(f) Up-Down (Anderson et al., 2018) utilizes a bottom-up attention mechanism to capture image details, combined with top-down contextual information, to generate accurate and expressive captions.

(g) Transformer (Vaswani, 2017) adopts the standard Transformer encoder–decoder architecture, effectively achieving image caption generation.

(h) AoANet (Huang et al., 2019) employs an attention-on-attention mechanism to establish relationships between image features. Its decoder includes an LSTM and an attention-on-attention module.

(i) M2 Transformer (Cornia et al., 2020) embeds prior knowledge of relationships between objects into self-attention in both the encoder and decoder modules, each consisting of three identical layers.

(j) DLCT (Luo et al., 2021) leverages a dual-way self-attention module to explore intrinsic characteristics of features and introduces a comprehensive relation attention module to embed geometric information.

(k) RSTNet (Zhang et al., 2021) adopts a grid-augmented module that enhances visual representations with relative geometry features. It also incorporates a BERT-based language model and an adaptive attention module for dynamic cue measurement in word prediction using a Transformer decoder.

(l) S2 Transformer (Zeng et al., 2022) utilizes learnable clustered implicit learning pseudo-regions in a spatial-aware pseudo-supervised module to improve caption accuracy.

(m) LSTNet (Ma et al., 2023) adopts locality-sensitive attention and locality-sensitive fusion to model relationships between individual grids and their neighborhoods, enhancing the interaction and fusion of information to improve caption accuracy.

(n) MDSANet (Ji et al., 2023) introduces distance-sensitive self-attention and multi-branch self-attention to optimize self-attention networks for image captioning.

The experimental results are presented in Table 4. Our proposed UICM-SOFF demonstrates outstanding performance across all evaluation metrics, particularly excelling in BLEU-1, BLEU-2, BLEU-3, BLEU-4, CIDEr, SPICE, and S_m achieving scores of 0.7249, 0.5980, 0.4920, 0.3953, 1.2831, 0.2368, and 0.6415, respectively. This indicates a significant advantage of UICM-SOFF over other methods in terms of both precision and diversity in generating underwater image captions. Although UICM-SOFF does not reach the absolute optimal levels in the METEOR and ROUGE-L metrics (0.2988 and 0.5888, respectively), it still maintains a competitive position, demonstrating its strong capability in capturing the semantic richness and overall coherence of underwater image captions.

Compared with specific models, it is evident that performance significantly improves with the integration of self-attention mechanisms (e.g., Soft-Att, Transformer, etc.), underscoring their crucial role in understanding image content and generating captions. The Up-Down model further enhances these mechanisms, yielding notable improvements in BLEU and CIDEr scores. However, in direct comparison with these models, UICM-SOFF stands out among all attention-based methods due to its novel structure and optimization. Particularly noteworthy is UICM-SOFF's exceptional performance in the SPICE metric, achieving a score of 0.2368, highlighting its superior abilities in underwater scene interpretation. Although recent advanced models such as AoANet, M2 Transformer, DLCT, RSTNet, S2 Transformer, LSTNet, and MDSANet have shown some performance improvements, none surpasses our proposed UICM-SOFF comprehensively. While the S2 Transformer performs better than UICM-SOFF in METEOR and ROUGE-L metrics, there are still noticeable gaps in BLEU-1 to BLEU-4, CIDEr, and SPICE metrics.

In summary, the experimental results clearly demonstrate the superior performance of UICM-SOFF in underwater image captioning. The captions exhibit high accuracy and diversity, showcasing its exceptional ability to comprehend complex underwater scenes and maintain semantic consistency.

6.3. Ablation experiments

Experimental Evaluation of Different Weight Initialization Methods: We evaluate the performance of the UICM-SOFF based on different weight initialization methods on the underwater image captioning dataset using visual feature extractors of typical CNNs, namely VggNet-13 (Simonyan and Zisserman, 2014), GoogleNet (Szegedy et al., 2015), DenseNet (Huang et al., 2017), and ResNeXt (Xie et al., 2017). These weight initialization methods include random weight initialization, pre-trained weight initialization based on the ImageNet dataset, and meta-weight initialization based on the degraded image dataset. The experimental results are presented in Table 5.

Compared with random weight initialization, utilizing pre-trained weight initialization has already improved the performance of each

Table 4
Experimental results of our proposed UICM-SOFF and some existing image captioning models on the underwater image captioning dataset. B-N, M, R-L, C, and S denote BLEU-N, METEOR, ROUGE-L, CIDEr, and SPICE, respectively. The best results are presented in bold, and the second-best results are underlined.

Model	B-1 ↑	B-2 ↑	B-3 ↑	B-4 ↑	M ↑	R-L ↑	C ↑	S ↑	S_m ↑
NIC (Vinyals et al., 2015)	0.6551	0.5183	0.4161	0.3275	0.2550	0.5323	0.9229	0.2037	0.5094
Soft-Att (Xu et al., 2015)	0.6893	0.5544	0.4466	0.3543	0.2736	0.5557	1.0137	0.2137	0.5493
Bi-LSTM (Wang et al., 2016)	0.6813	0.5554	0.4534	0.3661	0.2761	0.5661	1.0747	0.2163	0.5707
Spatial-Att (Lu et al., 2017)	0.6772	0.5509	0.4414	0.3436	0.2639	0.5479	0.9670	0.2081	0.5306
Adaptive-Att (Lu et al., 2017)	0.6811	0.5624	0.4595	0.3650	0.2680	0.5535	0.9656	0.2118	0.5380
Transformer (Vaswani, 2017)	0.7030	0.5656	0.4496	0.3441	0.2797	0.5663	0.9923	0.2089	0.5456
Up-Down (Anderson et al., 2018)	0.7064	0.5807	0.4653	0.3585	0.2775	0.5615	1.1680	0.2114	0.5914
AoANet (Huang et al., 2019)	0.7075	0.5667	0.4498	0.3541	0.2965	0.5847	1.0760	0.2255	0.5778
M2 Transformer (Cornia et al., 2020)	0.6921	0.5610	0.4540	0.3630	0.2876	0.5745	1.1284	0.2229	0.5884
DLCT (Luo et al., 2021)	0.7012	0.5718	0.4615	0.3634	0.2945	0.5813	1.1777	0.2261	0.6042
RSTNet (Zhang et al., 2021)	0.7127	0.5804	0.4693	0.3730	0.2964	0.5859	1.1722	0.2259	0.6069
S2 Transformer (Zeng et al., 2022)	0.7222	0.5958	0.4877	0.3938	**0.3028**	**0.5944**	1.2481	0.2311	0.6348
LSTNet (Ma et al., 2023)	0.7173	0.5873	0.4792	0.3834	0.2978	0.5885	1.2104	0.2309	0.6200
MDSANet (Ji et al., 2023)	0.7124	0.5861	0.4769	0.3808	0.3027	0.5883	1.2124	0.2367	0.6210
UICM-SOFF	**0.7249**	**0.5980**	**0.4920**	**0.3953**	0.2988	0.5888	**1.2831**	**0.2368**	**0.6415**

Table 5
Experimental results of UICM-SOFF subject to different weight initialization methods, i.e., random weight initialization (RWI), pre-trained weight initialization (PWI) based on the ImageNet dataset, and meta-weight initialization (MWI) based on the degraded image dataset. "Backbone" refers to the visual feature extractors from various CNNs. The bold black figures indicate the best results.

Backbone	Weight initialization	B-1 ↑	B-2↑	B-3 ↑	B-4 ↑	M ↑	R-L ↑	C ↑	S ↑	S_m ↑
VggNet-13	RWI	0.6162	0.4663	0.3524	0.2543	0.2386	0.4834	0.6645	0.1699	0.4102
	PWI	0.7063	0.5756	0.4662	0.3690	0.2919	**0.5803**	1.1943	0.2266	0.6089
	MWI	**0.7078**	**0.5843**	**0.4805**	**0.3864**	**0.2966**	0.5781	**1.2201**	**0.2324**	**0.6203**
GoogleNet	RWI	0.6157	0.4789	0.3640	0.2604	0.2419	0.4856	0.6656	0.1733	0.4134
	PWI	0.7048	0.5706	0.4585	0.3574	0.2833	0.5697	1.0999	0.2227	0.5776
	MWI	**0.7163**	**0.5848**	**0.4740**	**0.3746**	**0.2942**	**0.5818**	**1.1918**	**0.2276**	**0.6106**
DenseNet	RWI	0.6077	0.4622	0.3457	0.2444	0.2321	0.4717	0.6240	0.1641	0.3931
	PWI	0.7166	0.5849	0.4763	0.3784	0.2921	0.5814	1.2169	0.2328	0.6172
	MWI	**0.7289**	**0.5982**	**0.4908**	**0.3907**	**0.3013**	**0.5891**	**1.2610**	**0.2362**	**0.6355**
ResNeXt	RWI	0.5859	0.4440	0.3280	0.2291	0.2292	0.4621	0.5219	0.1486	0.3606
	PWI	0.7088	0.5762	0.4683	0.3761	0.2929	0.5874	1.1935	0.2287	0.6125
	MWI	**0.7249**	**0.5980**	**0.4920**	**0.3953**	**0.2988**	**0.5888**	**1.2831**	**0.2368**	**0.6415**

Table 6
Experimental results of UICM-SOFF by using end-to-end and non-end-to-end training methods. The bold blue figures indicate the best results.

Backbone	Training methods	B-1 ↑	B-2 ↑	B-3 ↑	B-4 ↑	M ↑	R-L ↑	C ↑	S ↑	S_m ↑
VggNet-13	Non-end-to-end	0.6954	0.5594	0.4502	0.3543	0.2843	0.5624	1.1125	0.2236	0.5783
	End-to-end	**0.7078**	**0.5843**	**0.4805**	**0.3864**	**0.2966**	**0.5781**	**1.2201**	**0.2324**	**0.6203**
GoogleNet	Non-end-to-end	0.6888	0.5553	0.4473	0.3550	0.2802	0.5616	1.0397	0.2142	0.5591
	End-to-end	**0.7163**	**0.5848**	**0.4740**	**0.3746**	**0.2942**	**0.5818**	**1.1918**	**0.2276**	**0.6106**
DenseNet	Non-end-to-end	0.7143	0.5807	0.4689	0.3687	0.2860	0.5732	1.1623	0.2239	0.5975
	End-to-end	**0.7289**	**0.5982**	**0.4908**	**0.3907**	**0.3013**	**0.5891**	**1.2610**	**0.2362**	**0.6355**
ResNeXt	Non-end-to-end	0.6994	0.5647	0.4560	0.3596	0.2847	0.5679	1.1560	0.2216	0.5920
	End-to-end	**0.7249**	**0.5980**	**0.4920**	**0.3953**	**0.2988**	**0.5888**	**1.2831**	**0.2368**	**0.6415**

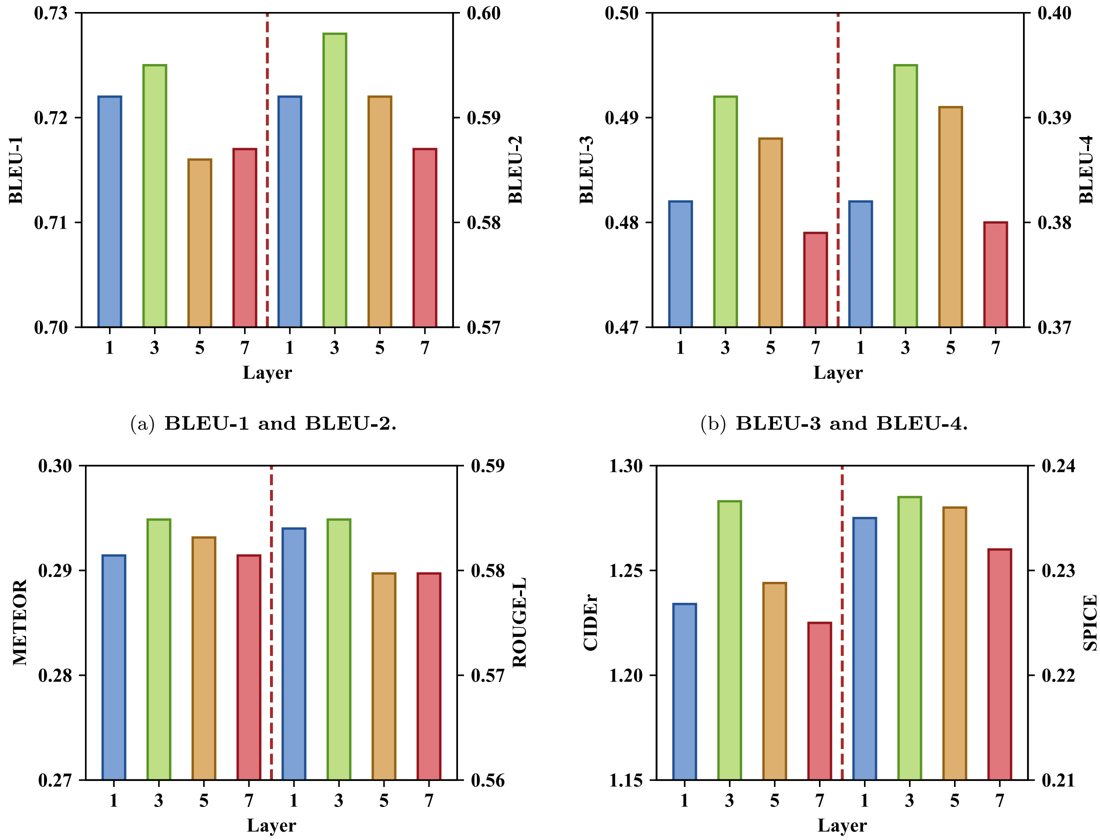
model. However, incorporating meta-weight initialization further optimizes the visual feature extractors, enhancing their feature extraction capability across various scenes and effectively adapting to the distinctive conditions of underwater environments. Specifically, for VggNet-13, GoogleNet, DenseNet, and ResNeXt, under the condition of initialization by meta-weights, BLEU-4 scores increased by 0.1321, 0.1142, 0.1463, and 0.1662, respectively, compared with random weight initialization, accompanied by a significant increase in CIDEr scores.

In summary, we propose a model pre-trained on degraded images that generates meta-weights to initialize UICM-SOFF, enhancing the performance of models in the underwater image captioning task. Compared with random weight initialization or pre-trained weight initialization based on the ImageNet dataset, this method enables better capture and adaptation of underwater image features, yielding more accurate underwater image captions.

Experimental Evaluation of the UICM-SOFF Using End-to-End and Non-End-to-End Training Methods: Based on whether the visual feature extractor of UICM-SOFF is included in training, we categorize the training methods into end-to-end (included) and non-end-to-end (not included) methods. The experiment involves four CNNs as mentioned earlier, and the results in Table 6 demonstrate that the end-to-end method outperforms the non-end-to-end method across various evaluation metrics.

Specifically, significant improvements are observed in scores for BLEU-1 to BLEU-4, METEOR, ROUGE-L, CIDEr, and SPICE metrics. For example, using VggNet-13, the end-to-end method results in a 0.0321 increase in BLEU-4 score and a 0.1076 improvement in CIDEr compared with the non-end-to-end method. Similar results are observed in the other three network architectures, indicating that the end-to-end method can notably enhance the performance of models based on various CNN architectures in the underwater image captioning task. The end-to-end method not only simplifies the process but also achieves greater improvements in key metrics compared with the non-end-to-end method, demonstrating its superiority in underwater image captioning.

Experimental Evaluation of Different Numbers of Encoder Layers: We investigate the impact of varying number of scene-object feature fusion encoder layers in UICM-SOFF on the underwater image



(a) **BLEU-1 and BLEU-2.**

(b) **BLEU-3 and BLEU-4.**

(c) **METEOR and ROUGE-L.**

(d) **CIDEr and SPICE.**

Fig. 7. Experimental results of UICM-SOFF with respect to different numbers of scene-object feature fusion encoder layers.

captioning. The experimental results in Fig. 7 present performance across various evaluation metrics under different encoder layer configurations. The analysis reveals that increasing the number of encoder layers from one to three leads to significant improvements across all metrics, with the three-layer configuration demonstrating superior performance. This indicates that moderately increasing the network depth can greatly enhance the model's ability to understand underwater images and generate accurate captions. However, increasing the number of encoder layers further to five and seven leads to slight decreases in certain metrics, such as BLEU-4, METEOR, and SPICE. This suggests that increasing network depth beyond three layers can introduce challenges, such as overfitting or increased model complexity, without a proportional increase in performance.

Overall, the experimental results demonstrate that utilizing three scene-object feature fusion encoder layers achieves the optimal balance between model complexity and performance, establishing it as the most effective configuration for underwater image captioning with UICM-SOFF.

Experimental Evaluation of the Threshold for the Number of Detected Objects: We investigate the impact of the threshold for the number of detected objects in UICM-SOFF on the underwater image captioning. The results in Table 7 present the performance across various evaluation metrics under different thresholds. Increasing the threshold T from 10 to 50 results in diverse improvements across multiple evaluation metrics, including BLEU, METEOR, ROUGE-L, CIDEr, SPICE, and S_m, with optimal performance achieved at $T = 50$. However, further increasing T to encompass all detected objects does not sustain this upward trend. Instead, certain metrics exhibit a decline, suggesting that a higher number of detected objects does not necessarily yield superior results. Therefore, based on these empirical findings and considering both performance and computational efficiency, we set $T = 30$ as the threshold for the number of detected objects within UICM-SOFF.

6.4. Qualitative analysis

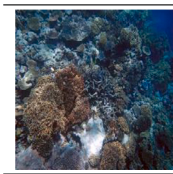
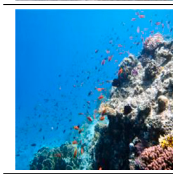
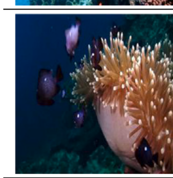
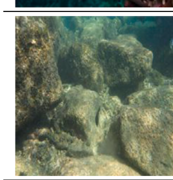
We perform a qualitative analysis comparing the results of our proposed UICM-SOFF model with those of a standard Transformer-based image captioning model (Vaswani, 2017). We use ground-truth reference captions (GT1 to GT3) as benchmarks to assess the quality of the generated captions. Table 8 shows various underwater scenes where our model consistently outperforms the standard Transformer-based image captioning model in several aspects: accurately capturing details, emphasizing the features of main objects, and integrating environmental context to generate captions that closely align with the ground-truth reference captions.

For example, in the first row of Table 8, our model accurately captures the diversity of shapes in coral reef scenes on the seabed, while the standard Transformer-based image captioning model incorrectly recognizes the objects as "many small fish". In the second row of Table 8, our model precisely describes the color feature of the object "small fish" in underwater images as "red" and vividly depicts the scene of small red fish swimming in the "blue water" near the coral reef. Moreover, in the third row of Table 8, our model correctly recognizes the objects as "purple black fish", whereas the standard Transformer-based image captioning model mistakenly recognizes them as "clownfish". Similarly, in the fourth row of Table 8, our model provides a more accurate caption, whereas the Transformer-based image captioning model describes objects, "many small fish", that are not present in the underwater image. Overall, our proposed UICM-SOFF model excels over the standard Transformer-based image captioning model in capturing details, accuracy, and consistency.

Table 7
Experimental results of UICM-SOFF with respect to different thresholds of the number of detected objects. “All” represents containing all detected objects. The best results are presented in bold, and the second-best results are underlined.

Threshold	B-1 ↑	B-2 ↑	B-3 ↑	B-4 ↑	M ↑	R-L ↑	C ↑	S ↑	S_m ↑
10	0.7199	0.5859	0.4749	0.3727	0.2999	0.5874	1.2663	0.2341	0.6316
30	0.7249	0.5980	0.4920	0.3953	0.2988	0.5888	1.2831	0.2368	0.6415
50	0.7335	**0.6102**	**0.5030**	**0.4040**	**0.3028**	**0.6035**	**1.2969**	**0.2416**	**0.6518**
All	**0.7340**	0.6012	0.4909	0.3893	0.2991	0.5954	1.2734	0.2358	0.6393

Table 8
Qualitative analysis of underwater image captioning results by using UICM-SOFF and a standard Transformer-based image captioning model. The blue and purple colors are used to highlight the main distinctions in the generated underwater image captions.

Underwater image	Underwater image caption
	GT-1: There are corals of various colors and shapes on the seabed. GT-2: Many corals grow on the reefs at the seafloor. GT-3: There are corals of various colors and shapes on the reefs on the seabed. Transformer: Many small fish swim near the coral reef. UICM-SOFF: There are many corals of different shapes growing on the seabed.
	GT-1: A group of small red fish swim in the blue seawater next to the reef. GT-2: Lots of small red fish swimming in clear blue water around coral reef. GT-3: Many small red fish swim around the coral reef on the seabed. Transformer: Many small fish swim near the coral reef. UICM-SOFF: Many small red fish swim in the blue water near the coral reef.
	GT-1: Several deep purple small fish play around near a sea anemone. GT-2: There are several small fish with deep purple coloration playing near a sea anemone. GT-3: Several deep purple small fish swim near the sea anemone. Transformer: A clownfish swims in the sea anemone. UICM-SOFF: Some small purple black fish swim around the sea anemone.
	GT-1: The clear water has many rocks of different shapes. GT-2: There are many rocks of different sizes on the seabed. GT-3: An underwater scene with rocks of various sizes and shapes. Transformer: Many small fish swim near the rocks on the seabed. UICM-SOFF: There are many rocks of different sizes and shapes on the seabed.

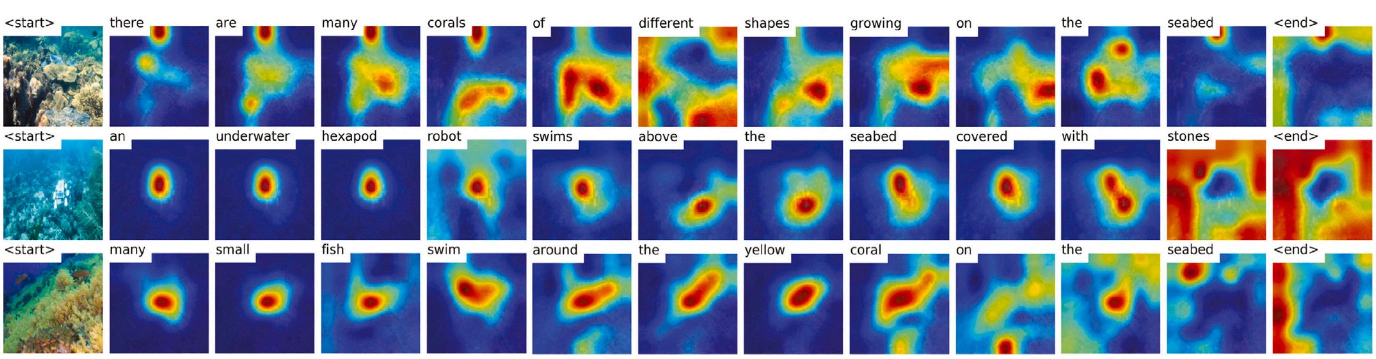


Fig. 8. Visual attention maps of UICM-SOFF on exemplary underwater images. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

6.5. Attention visualization

In this subsection, we present attention visualization results of activation maps from UICM-SOFF to illustrate the importance of visual features in generating the final output words. Specifically, we select the multi-head attention weights of the final layer of the Transformer decoder for visualization.

For the exemplary underwater image in the first row of Fig. 8, the caption generated by UICM-SOFF, “there are many corals of different shapes growing on the seabed”, accurately describes the object information “many corals” and their positional information “on the seabed”. For the exemplary underwater image in the second row, “an underwater hexapod robot swims above a seabed covered with stones”, the caption clearly highlights the robot’s unique hexapod structure, and the activation maps focus on the areas corresponding to “robot” and “stones”. For the exemplary underwater image in the third row, "many small fish swim around the yellow coral on the seabed" accurately depicts the relationship between the object "small fish" and the scene "around the yellow coral on the seabed", providing a clear and detailed caption for the underwater image. Overall, the activation maps demonstrate that the UICM-SOFF model proposed in this study generates underwater image captions that describe effectively both the objects and scenes. Furthermore, the captions exhibit good coherence and natural language fluency.

7. Conclusion

To advance the evolution of underwater photogrammetry from basic "visual perception" to "semantic understanding", we have developed novel underwater image captioning technology to enhance our understanding of underwater environments. Specifically, we have proposed a novel meta-learning strategy that bridges the gap between natural and underwater image domains. Furthermore, we have developed an underwater image captioning model based on scene-object feature fusion. This model aims to yield comprehensive features for underwater image captioning by combining underwater scene features extracted using Meta-ResNeXtor with object features localized by YOLOv8. Additionally, we have constructed and released datasets related to underwater image captioning to train and evaluate our proposed model.

Despite the aforementioned advancements, the implementation process remains relatively complex due to the need for multiple training phases in the underwater image captioning model. In addition, the limited size of the underwater image captioning dataset affects the model's generalization ability. In future work, we will explore the use of fine-tuning mechanisms with large language models to simplify the training process. At the same time, we will develop a more comprehensive and diverse set of underwater image captioning datasets to improve the model's performance and adaptability in underwater image captioning tasks.

CRediT authorship contribution statement

Huanyu Li: Writing – original draft, Methodology, Investigation. **Hao Wang:** Writing – review & editing, Methodology. **Ying Zhang:** Writing – review & editing. **Li Li:** Writing – review & editing. **Peng Ren:** Writing – review & editing, Methodology, Investigation, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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